

Aadit Kannan

Project Portfolio

This portfolio includes my hardware, electrical, and software projects, along with some of my physics and matsci work on semiconductor thin-films.

Some of my projects are included here. For the full list, go to aaditkannan.com/projects.



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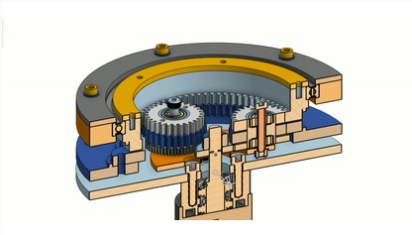
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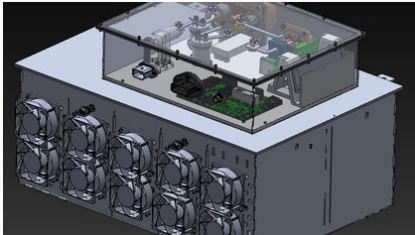
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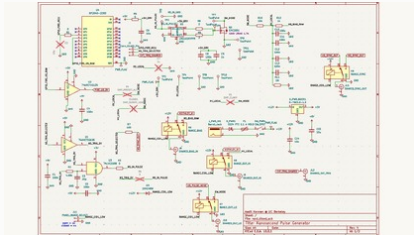
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Actuator section render



Accumulator attic CAD

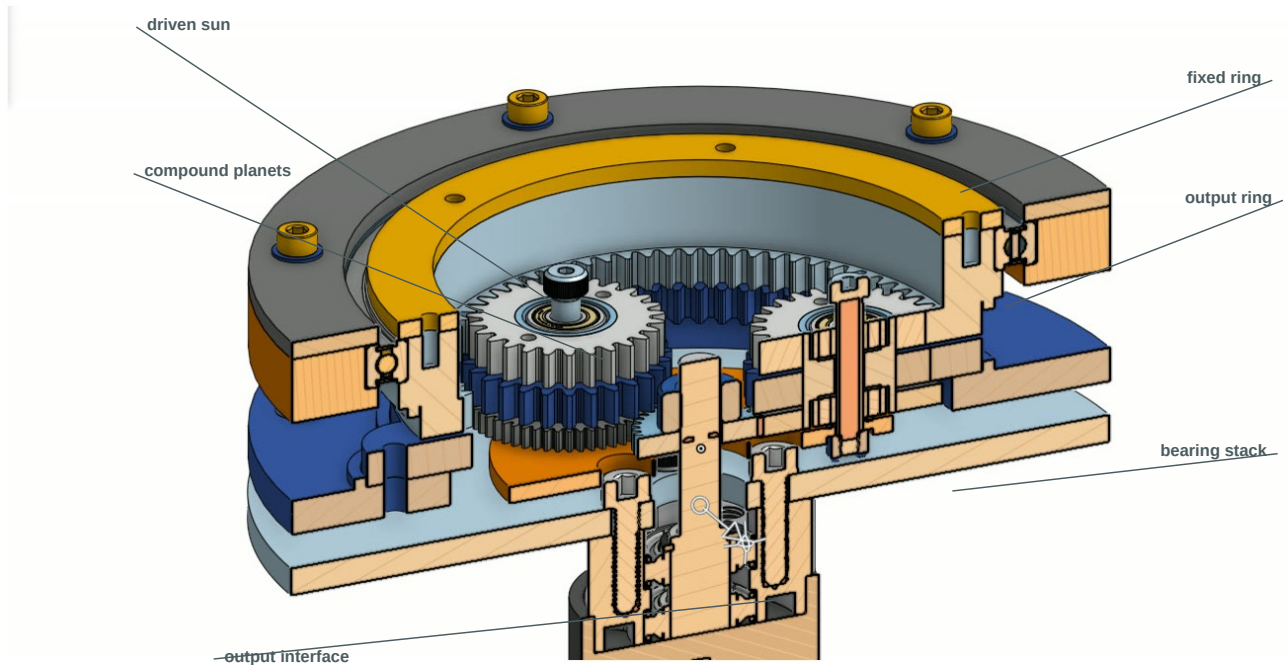


Pulse-generator schematic

Wolf from Robotic Actuator

Jun 2026 - Present / Robotics Hardware

Compact Wolf from-style actuator for humanoid-scale joints. The current build is about packaging a high-ratio gearbox into a small module and learning what has to change before the metal version.



CAD section: driven sun, compound planets, fixed ring, output ring, bearing stack, and output interface.

DESIGNED

- Full stackup: sun, compound planets, fixed ring, output ring, carrier, bearings, shafts, and output hub.
- One compact axial package instead of stacking multiple planetary stages.
- Split output ring, bolt access, dowel alignment, bearing support, and assembly order.
- Printed geometry first; metal revision planned around CNC aluminum structure and wire-EDM ring gears.

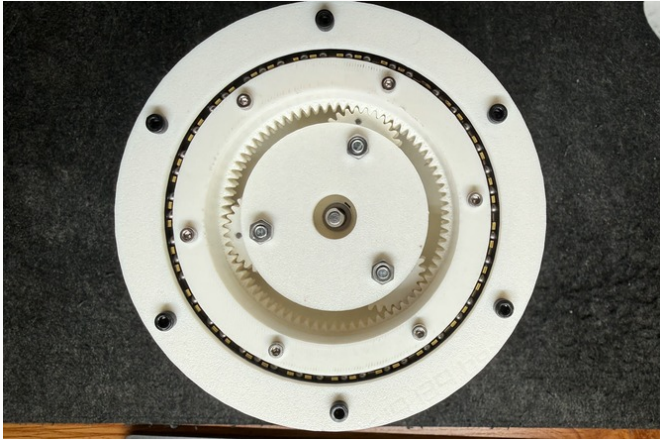
GEARBOX NOTES

- Current tooth counts give 555/11, about 50.45:1. I am also exploring a 125:1 variant.
- The ratio split matters because too much reduction in the ring differential gets lossy fast.
- Main things I am watching: ring mesh friction, planet bearing drag, load sharing, and rough printed teeth.
- Backdrivability is being treated as a friction problem, not as a hand-wavy claim.

Wolf from Robotic Actuator

Jun 2026 - Present / Robotics Hardware / Prototype + Next Revision

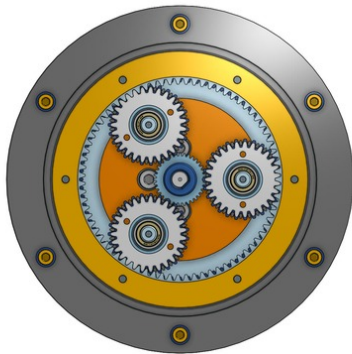
The printed parts are not meant to prove final performance. They are there to catch mesh, stack height, bearing, and assembly problems before I spend real money on the metal build.



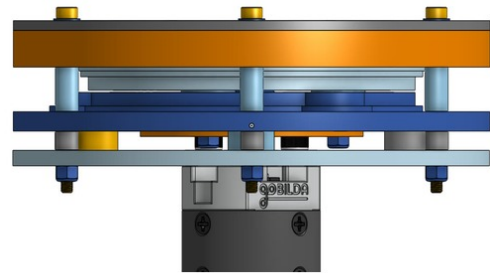
Printed prototype used for gear mesh, bearing placement, and stack-height checks.



Bench assembly and fit-checking before re-timing the actuator by hand.



Top view: three compound planets, output ring, fastener pattern, and center sun.



Side view: plate stack, motor interface, standoffs, and axial package.

CURRENT PROTOTYPE

- Printed and fit-checked the actuator hardware for mesh, assembly access, and stack alignment.
- Working on split-gear preload to reduce backlash; no final backlash number yet.
- Treating printed gears as geometry prototypes, not load-rated actuator parts.

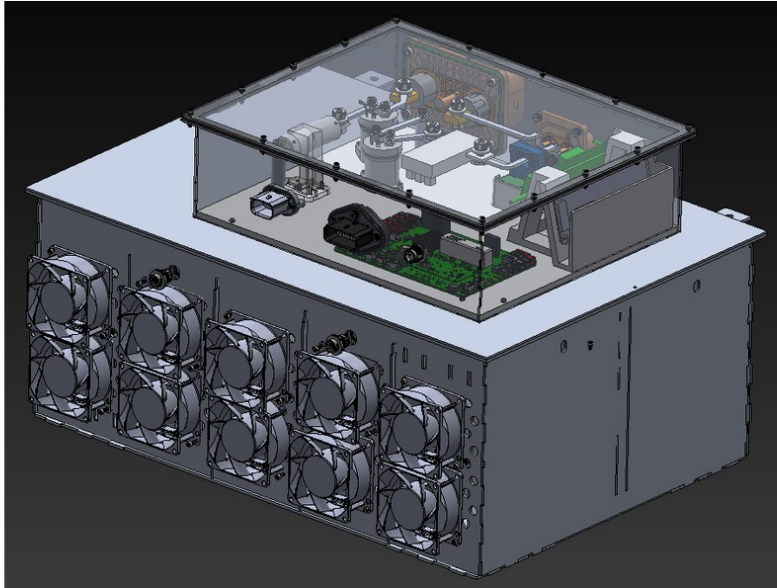
NEXT REVISION

- Use KISSsoft to clean up macro-geometry, profile shifts, and tip relief.
- Move to CNC aluminum structure, wire-EDM hardened-steel rings, hardened pins, and needle bearings.
- Test breakaway torque, backdrive feel, efficiency, and load behavior after the metal revision exists.

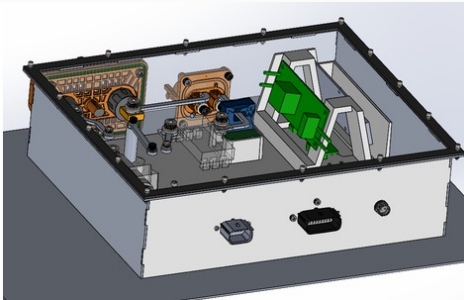
Formula Electric 588V Accumulator

Sep 2025 - Present / High-Voltage Packaging

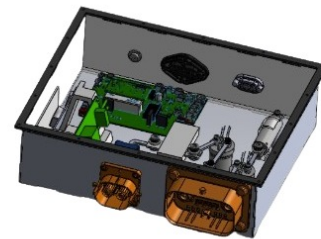
Accumulator attic and enclosure work for Berkeley Formula Electric. The hard part is not one bracket; it is making HV electronics, busbars, insulation, cooling, water sealing, and service access all fit in the same small volume.



588V accumulator CAD showing the HV electronics attic, fan wall, service access, and enclosure packaging.



Electronics attic layout: board, connector, busbar, and mechanical interface packaging.



Internal attic CAD view with board/connector access and enclosure clearances.

BUILT

- Packaged the HV attic: boards, busbars, fans, fasteners, service interfaces, and inspection access.
- Cut attic mass by 18% while keeping clearances, cooling paths, and fastener access workable.
- Designed around a 588V pack, water sealing, dielectric isolation, and SES/ESF review needs.

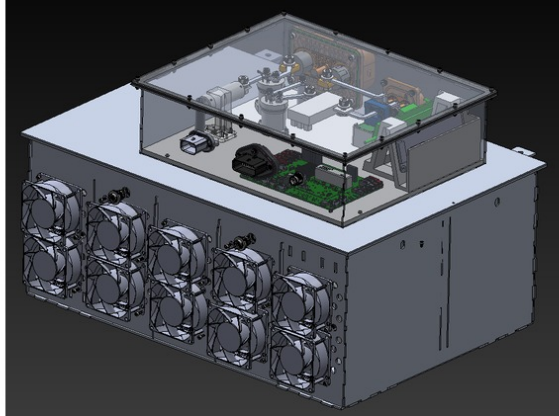
CONSTRAINTS

- Busbar routing for 80A peak discharge without making the attic impossible to service.
- FSAE load cases, HV isolation, rain-test sealing, airflow, and board access all compete for the same space.
- Transparent/inspectable surfaces matter because serviceability is part of safety.

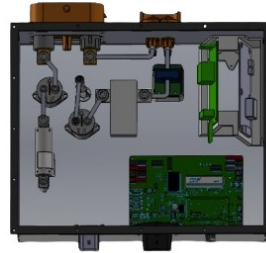
Formula Electric 588V Accumulator

Sep 2025 - Present / High-Voltage Packaging / Materials + Validation

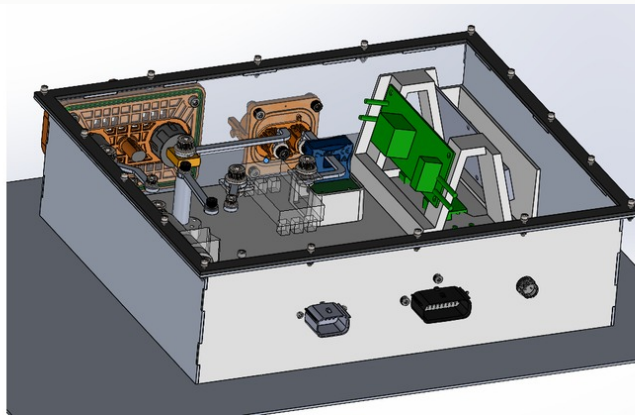
This page is the less flashy side of the accumulator: materials, manufacturing methods, and the checks that make the CAD believable.



Overall accumulator CAD context: attic, enclosure, service access, and internal packaging.



Top-down attic CAD view used to check electronics layout and access.



Detailed accumulator subassembly view for board, connector, and internal support packaging.

MANUFACTURING PLAN

- Laser-cut aluminum panels, TIG welded structure, waterjet polycarbonate floor, and waterjet neoprene gasket.
- Polycarbonate floor gives electrical isolation and visual inspection without tearing the pack apart.
- Nomex 410 for dielectric strength, temperature margin, and flame resistance near HV components.

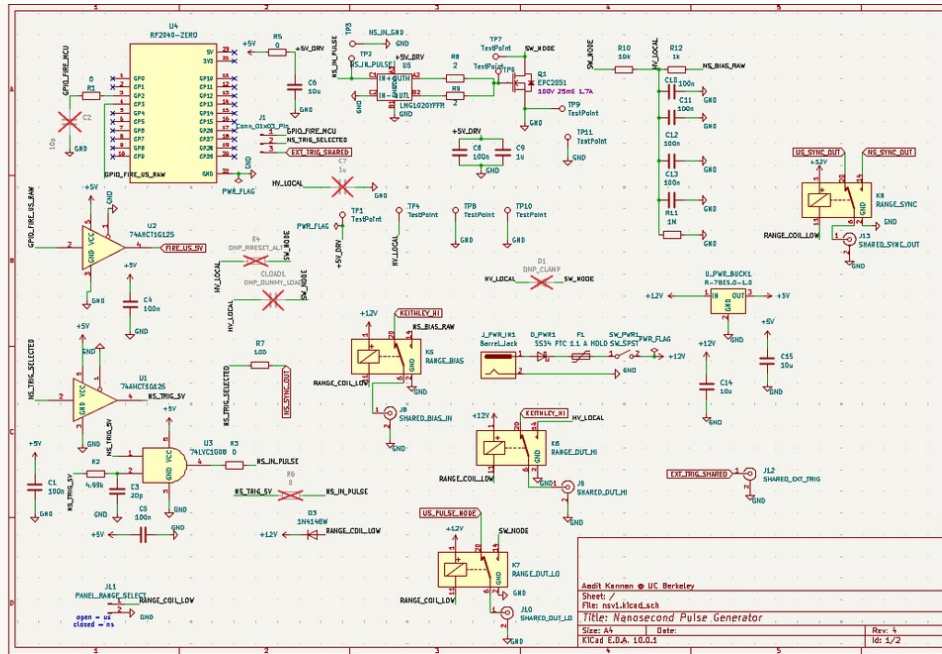
CHECKS

- FEA against 40g lateral, 40g longitudinal, and 20g vertical load cases.
- Checked gasket compression, water ingress, dielectric margin, service access, fastener access, and fan airflow together.
- Target gates include 1500V AC dielectric testing and IP65-style water ingress testing before competition use.

Nanosecond Pulse Generator

May 2026 - Present / Research Hardware / PCB

Board for fast electrical testing of BiFeO₃ devices in Ramesh Lab. The current design is moving from a rough microsecond pulser toward cleaner ns/us triggering, grounded coax/BNC I/O, and repeatable scope/SMU measurements.



Current nanosecond pulse-generator schematic. Shown as the main artifact because it is the current board revision.

BENCH INTERFACE

Python/RP2040 trigger -> driver and switching stage -> coax/BNC output -> device or probe station -> oscilloscope and source-measure capture.

DESIGNED

- Dual-range pulse architecture around current BFO testing near 30V, with future lower-voltage sweeps planned.
- GaN switching path, trigger/sync routing, grounded lab I/O, and source-measure bias interface.
- Layout organized around probe-station use, dummy-load bring-up, and scope visibility.

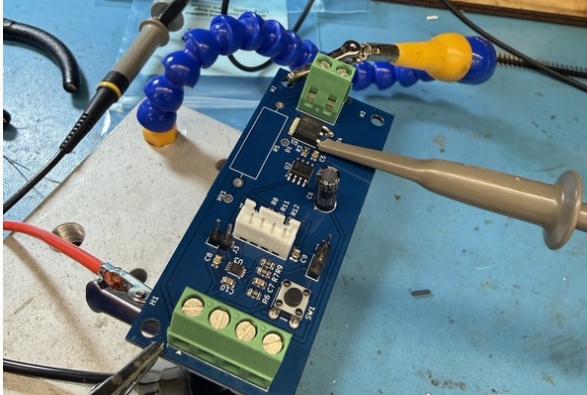
DEVICE-SIDE NOTES

- What matters is the voltage that actually lands across the BFO device, not just the connector setting.
- Measured current can mix polarization switching, leakage, dielectric capacitance, and fixture parasitics.
- No final rise-time, jitter, device result, or amplitude-stability claim yet; bring-up is still in progress.

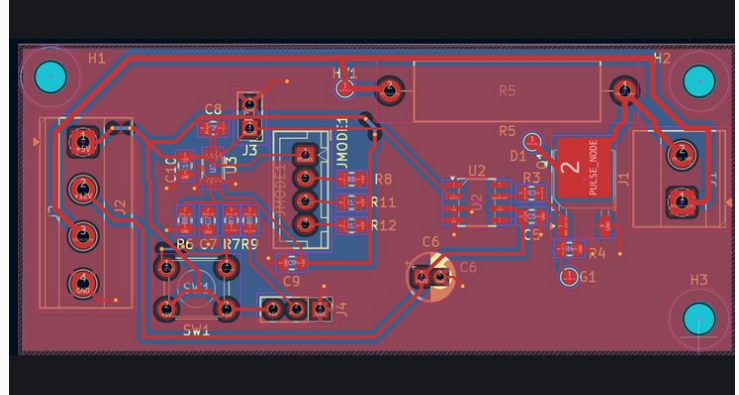
Nanosecond Pulse Generator

May 2026 - Present / Research Hardware / Board + V1 Learning

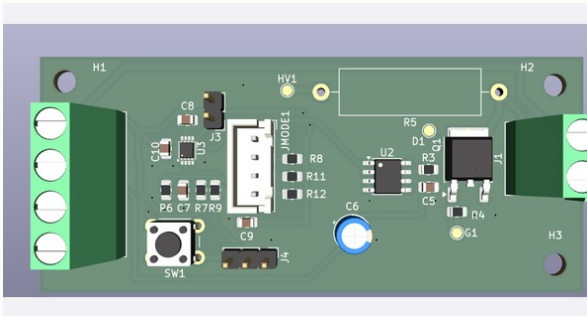
V1 was useful because it showed what was annoying on the bench. The current board keeps the parts that worked and fixes the triggering, lab I/O, grounding, and measurement flow.



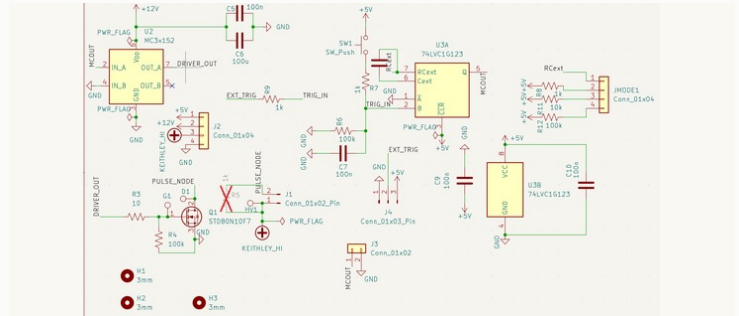
V1 microsecond board during bench probing. Useful as a learning artifact, not the current ns revision.



V1 PCB layout used to inspect trace routing, connector placement, and returns.



V1 rendered PCB for physical integration reference.



V1 microsecond schematic, included only as a predecessor to the current ns/us board.

V1 EXPOSED

- Clip-lead power and weak lab I/O made the first board awkward around real bench equipment.
- Triggering needed to be tied into Python/RP2040 control, not treated as a loose pulse node.
- LTspice transient work helped with timing intuition, but the current PDF does not rely on the ugly screenshot.
- Grounding and connector strategy had to be part of the instrument, not a last-minute bench hack.

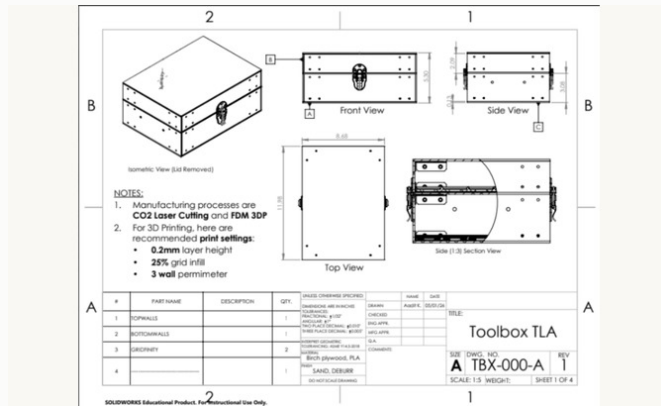
CURRENT REVISION ADDS

- Nanosecond and microsecond paths in one cleaner board.
- Coax/BNC-style outputs, shared sync/trigger handling, source-measure biasing, and clearer return paths.
- Bring-up order: rails, trigger logic, dummy loads, switching behavior, then device-side fixture measurements.

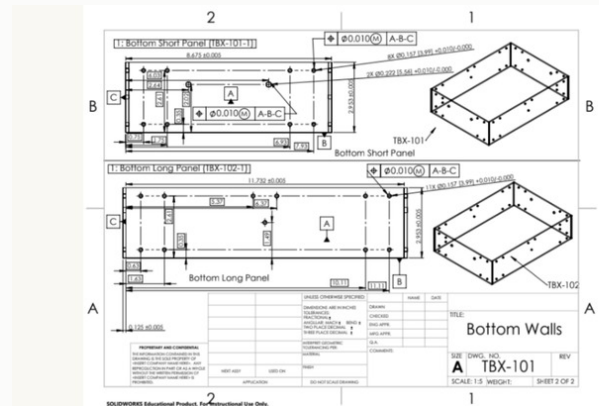
Custom Toolbox Design + Manufacturing

May 2026 - Jun 2026 / Mechanical Design

Toolbox project focused on the boring-but-important part of mechanical design: drawings, datums, tolerances, fabrication choices, and whether the thing actually fits after it is made.



Assembly drawing defining lid, latch, hinge, body, and insert interfaces.



Part drawing with datum references and positional tolerance callouts.



Prototype fit-up after fabrication with hinges, latches, handle, and fasteners.



Open prototype showing interior fit-up; final-version photos unavailable.

DESIGNED

- Production-style drawings using ASME Y14.5-2018 GD&T conventions.
- Datum scheme, position tolerances, fastener clearances, hinge/latch interfaces, and lid closure.
- Laser-cut plywood body with FDM/Gridfinity-style inserts and visible fastened joints.

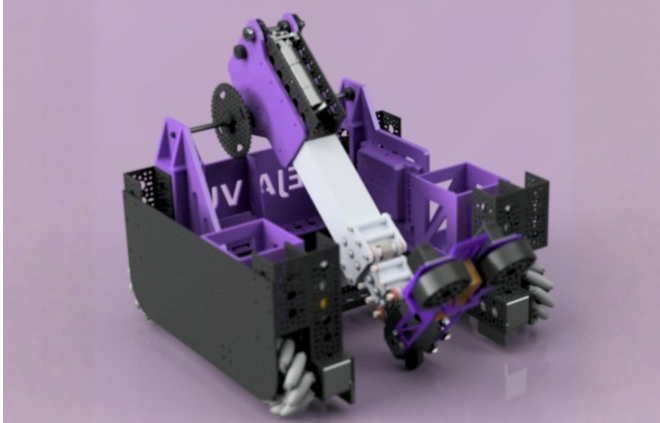
INSPECTED

- Fit-up checked with calipers, fasteners/gauge pins, squareness checks, lid closure, and hinge/latch alignment.
- The useful part of this project is drawing discipline and datum strategy, not pretending the box is more advanced than it is.
- Prototype photos are included honestly because final-version photos are not available.

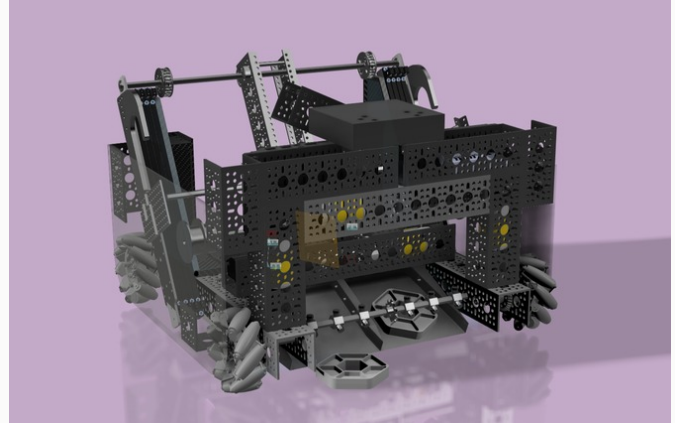
FIRST Robotics Hardware

Aug 2021 - Jun 2025 / Robotics Hardware / Controls

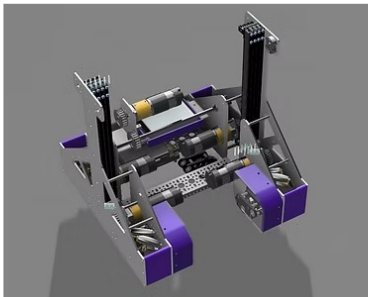
Four seasons of FTC hardware: 8 robots, 500+ part CAD assemblies, mechanism iteration under match constraints, Java controls, and three Robot Design Awards.



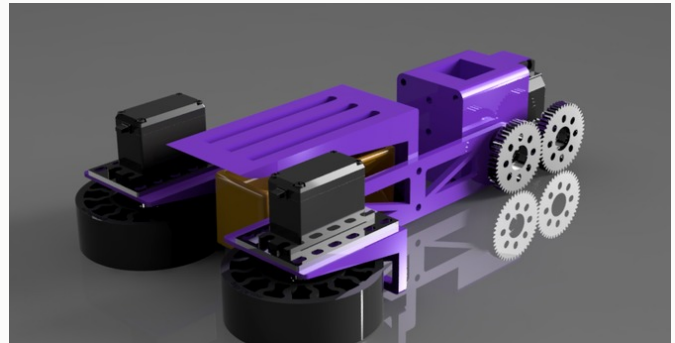
INTO THE DEEP robot CAD with drivetrain, active intake, transfer path, lift, and end effector.



CENTERSTAGE robot CAD assembly with pixel scoring mechanisms.



POWERPLAY robot CAD with lift and cone manipulation systems.



High-iteration intake subsystem CAD from Deja Vu INTO THE DEEP.

BUILT ACROSS SEASONS

- Mecanum drivetrains, active intakes, transfer paths, arms/lifts, claws, end effectors, hang mechanisms, and scoring systems.
- FEA-informed custom aluminum and printed components for stress, deflection, and mechanism clearance.
- Parts made through printing, laser-cut plates, CNC aluminum, and normal shop fabrication.

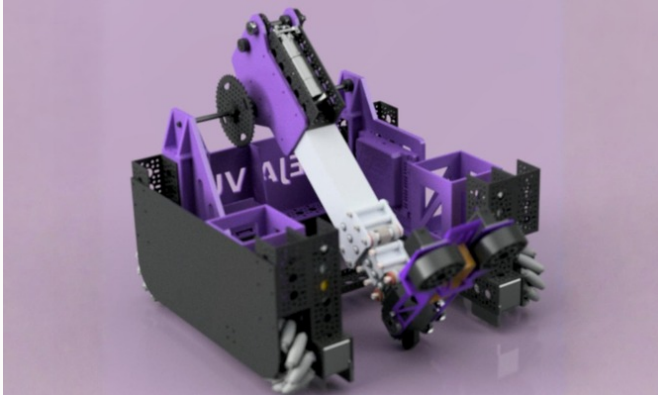
CONTROLS-AWARE DESIGN

- Java TeleOp with field-centric mecanum, encoder PID loops, mechanism sequencing, driver feedback, and autonomous routines.
- Hardware and driver code were tuned together because a mechanism that only works in CAD is not useful in a match.
- Mentored 30+ members in CAD and Java; this mattered because design quality depended on the build team actually understanding the robot.

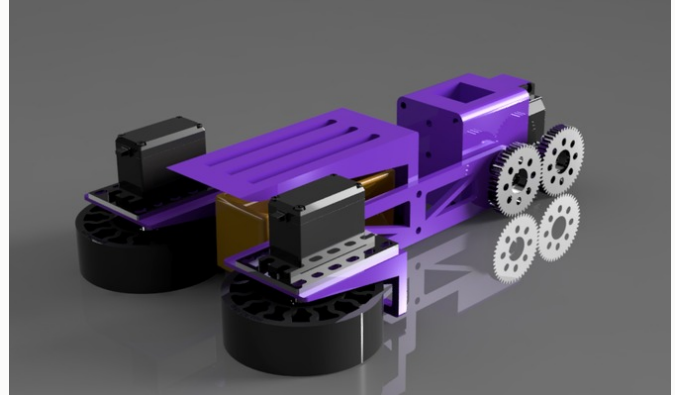
Deja Vu - INTO THE DEEP

Aug 2024 - Jun 2025 / FTC Robotics / Hardware Lead

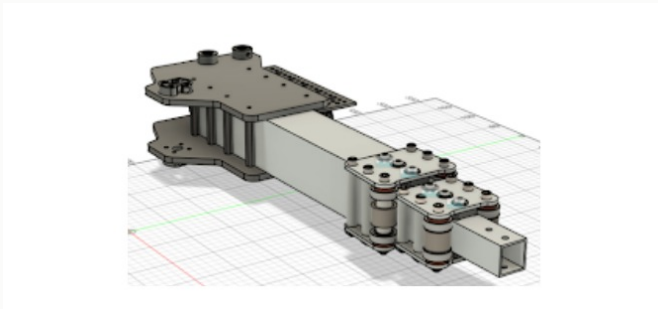
500+ part CAD robot for sample pickup, basket scoring, specimen scoring, and endgame hang. This was a full season of making mechanisms survive real driver practice and matches.



Full robot CAD with intake, transfer, lift, end effector, drivetrain, and hang packaging.



High-iteration intake mechanism CAD.



End-effector/linkage packaging detail.



Physical robot reference for mechanism packaging and service access.

DESIGNED

- Owned mechanical design, CAD management, and manufacturing coordination for FTC #13216.
- Four intake revisions: passive funnel, surgical-tubing roller, dual-stage roller, then a simpler high-speed roller geometry.
- Designed mecanum drivetrain, active intake, transfer pathway, arm/lift, end effector, and hang hardware.

TECHNICAL WORK

- FEA on arm pivot and hang brackets; checked stress, safety factor, and deflection against mechanism travel.
- Manufactured with Markforged nylon, Prusa prototypes, laser-cut polycarbonate/Delrin, CNC aluminum, and shop tools.
- Wrote Java TeleOp with field-centric drive, arm/lift PID, intake sequencing, and driver rumble feedback.

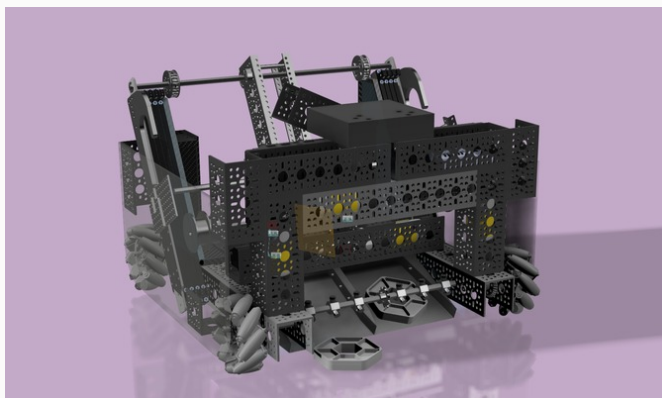
TOOLS

SolidWorks / FEA / Java / FTC SDK / PID Control / Mecanum Kinematics / Laser Cutting / Markforged

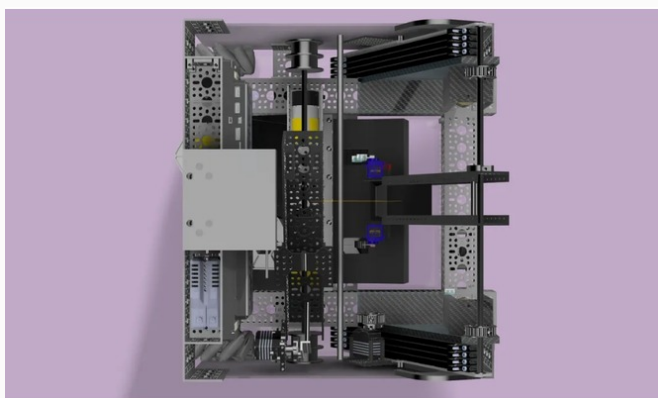
Deja Vu - CENTERSTAGE

Aug 2023 - Jun 2024 / FTC Robotics / Hardware Lead

CENTERSTAGE robot built around dual-pixel handling, backdrop scoring, vision alignment, and driver-controlled mechanism sequencing.



Full robot CAD assembly for CENTERSTAGE.



Robot/app or subsystem reference from the season.

DESIGNED

- Compliant TPU claw fingers, passive spring-loaded grip, and servo-actuated release for hexagonal pixels.
- Dual-pixel capacity to cut cycles to the backdrop; geometry had to grip without blocking the vision workflow.
- Robot packaging around mecanum drive, arm presets, claw actuation, and paper-airplane endgame launcher.

CONTROLS + RESULTS

- AprilTag-based autonomous alignment and camera-based pixel-position feedback.
- Java TeleOp with speed curves, arm presets, claw control, drone launcher arming, and intake/transfer coordination.
- Qualified for NorCal regional championship with very few mechanical failures in matches.

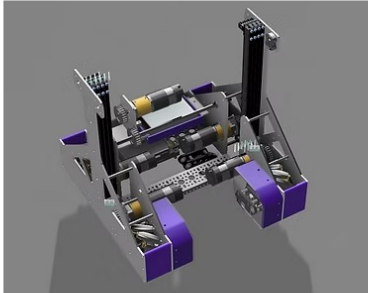
TOOLS

SolidWorks / Java / FTC SDK / AprilTags / OpenCV / PID Control / Mecanum Kinematics

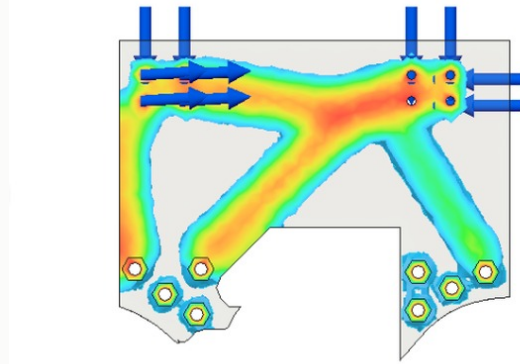
Zenith - POWERPLAY

Aug 2022 - Jun 2023 / FTC Robotics / Team Lead

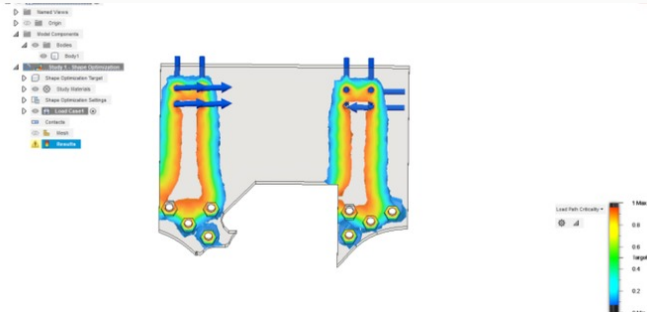
POWERPLAY robot built around a fast linear lift, cone-centering claw, mecanum drive, and autonomous positioning.



POWERPLAY robot CAD with lift and cone-scoring mechanism.



Mechanism/simulation reference for the season.



Additional CAD/simulation view.



Robot or subsystem image from Zenith.

BUILT

- Linear lift with 30+ inches of travel, sub-1.5s extension target, and about 200g carried load.
- String-rigged lift using dyneema and spool geometry balanced against motor torque and cycle time.
- Three claw revisions, landing on internal centering guides before gripping.

CONTROLS + LESSONS

- Java TeleOp with mecanum kinematics, lift PID, and speed scaling modes.
- Autonomous used OpenCV signal-sleeve detection, dead-wheel odometry, and IMU heading correction.
- The ZenLender parts spreadsheet started here and later turned into Inventory.

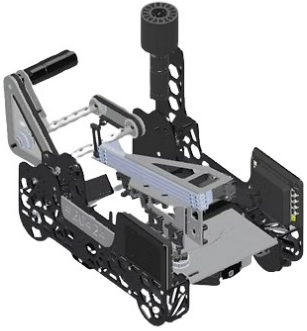
TOOLS

Onshape / Java / FTC SDK / OpenCV / Dead-Wheel Odometry / PID Control / String Rigging

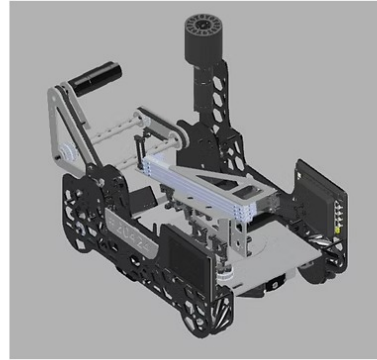
Zenith - FREIGHT FRENZY

Aug 2021 - Jun 2022 / FTC Robotics / First Season

First FTC season on Zenith #20424. This was where I learned CAD, basic manufacturing, Java control, and how robot subsystems actually go together.



Freight Frenzy robot image.



Freight Frenzy robot CAD/build reference.

LEARNED

- Onshape constraints, simple brackets, subassemblies, and complete robot packaging basics.
- Drill press, bandsaw, 3D printing, hole placement, print warping, and basic shop workflow.
- Motor-controller wiring and basic mechanism integration through a functioning FTC robot.

SOFTWARE

- Java gamepad input handling, arm/intake motor control, encoder-based autonomous moves, and carousel timing.
- Code was simple and hard-coded, but it established the foundation for later PID/state-machine work.
- Robot scored freight, delivered ducks, and parked successfully in matches.

TOOLS

Onshape / Java / FTC SDK / Encoder Autonomous / 3D Printing / Drill Press / Bandsaw

Serenity - FREIGHT FRENZY

Aug 2021 - Jun 2022 / FTC Robotics / Team Lead

Led a newer FTC team while also learning on Zenith. The robot was simple, but the team-process lessons mattered.



Serenity robot/team visual asset from the website.

BUILT

- Tank-drive robot with basic arm for freight placement, passive intake with compliant wheels, and carousel spinner.
- Design priority was reliability and match completion rather than maximum mechanism complexity.
- Competed in league meets with a functioning robot for a young team.

TEAM PROCESS

- Started weekly design reviews, shared CAD workspace conventions, manufacturing priority lists, and notebook assignments.
- Mentoring while still learning forced me to explain CAD, build, and competition decisions clearly.
- This was the first time I built team infrastructure, not only robot parts.

TOOLS

Onshape / Robot CAD / Manufacturing Planning / Engineering Notebook / Tank Drive / Carousel Mechanism

Construction Acclimation

Oct 2022 - Apr 2023 / Hardware / Data Logging

\$47 concrete-curing temperature/humidity probe and dashboard prototype. The goal was a cheaper way to log curing conditions and flag bad acclimation data.



Probe prototype hardware.



Electronics/enclosure or dashboard context.

BUILT

- Arduino Nano, DHT22, DS18B20, RTC, MicroSD logging, LiPo battery, fans, and PETG enclosure.
- Logged readings every 5 minutes with redundant temperature measurement.
- Client-side dashboard for CSV upload, trend plotting, out-of-range flags, and PDF reporting.

TESTED

- Compared against a calibrated Humboldt H-4210 commercial unit for 14 days.
- Temperature deviation about $\pm 0.8^{\circ}\text{C}$, humidity about $\pm 3.2\%$, and 4,028/4,032 readings captured.
- Presented to City of Palo Alto officials; certification, liability, and durability were still open issues.

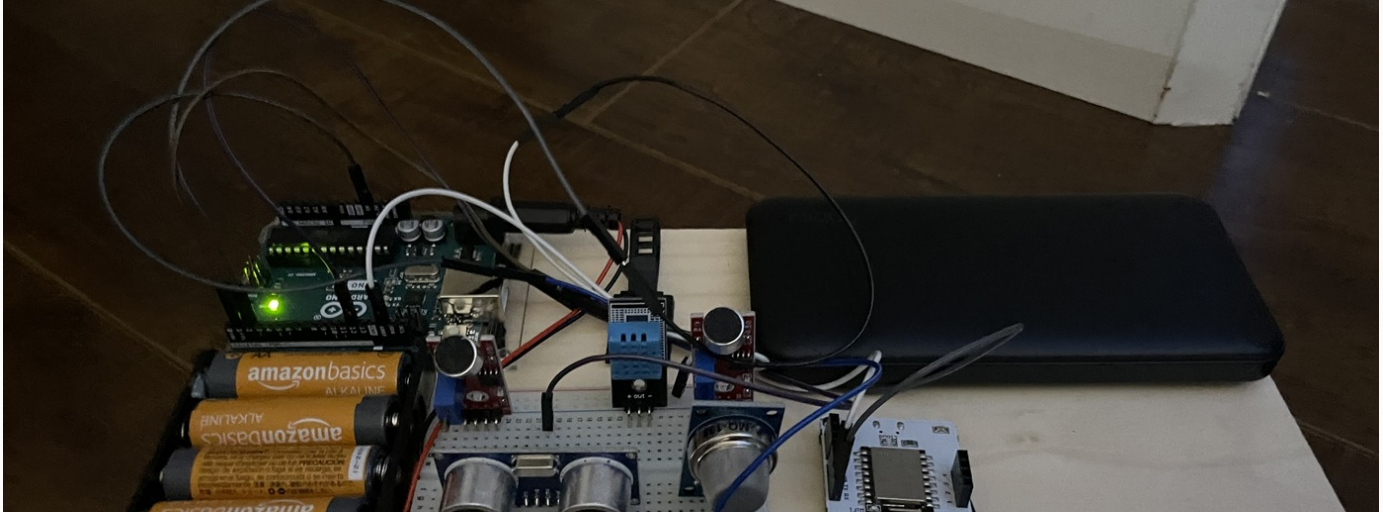
TOOLS

Arduino Nano / DHT22 / DS18B20 / RTC / MicroSD / Chart.js / PETG Enclosure

Friendly Fires

Sep 2021 - Mar 2022 / Sensor Hardware

Early sensor-hardware project for coal-seam fire detection: temperature, gas sensing, alerts, and a low-cost device concept.



Functional prototype electronics with sensors, power, microcontroller, and breadboard wiring.

BUILT

- Arduino-based sensor array for temperature, carbon monoxide, methane, and oxygen-depletion monitoring.
- Haptic, LED, and optional audio alerts intended for low-light mining environments.
- Rechargeable battery and low-power sleep-mode concept for a 12+ hour work shift.

PROTOTYPE STATUS

- Demonstrated controlled CO detection, temperature-gradient detection, and multi-sensor integration.
- Proof of concept only: no mining-environment calibration, false-positive study, dust/humidity durability, or regulatory approval.
- Included as an early sensor-integration project, not a deployable safety product.

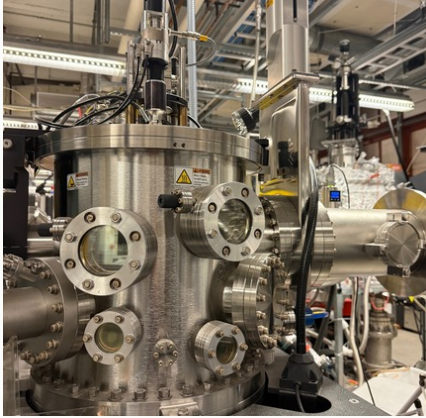
TOOLS

Arduino / CO Sensor / Methane Sensor / IR Temperature / Haptic Alerts / Sensor Integration

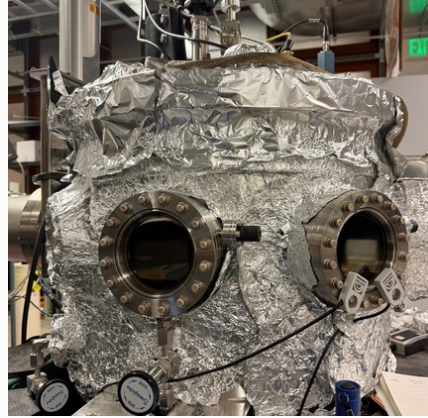
Ramesh Lab: Complex-Oxide Thin Films

Jan 2026 - Present / Research

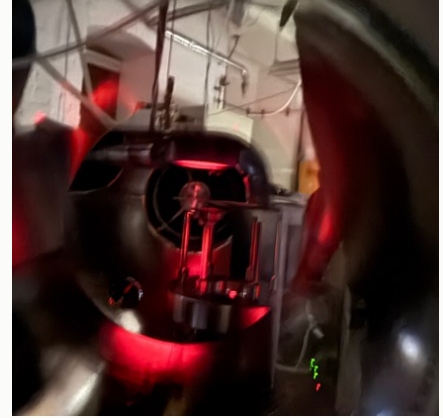
Research context for my Ramesh Lab work: BiFeO₃ and related complex-oxide thin films for memory/logic devices beyond conventional DRAM and CMOS.



Lab/growth workflow context.



Thin-film process and sample handling context.



Experimental hardware context from lab work.

LEARNING + GROWTH WORK

- Reading Lines and Glass for the physics base: polarization, domains, coercive fields, hysteresis, dielectric response, strain, and defects.
- Learning substrate prep, solvent cleaning, annealing, step-terrace control, PLD growth, and RHEED monitoring.
- Practicing the deposition workflow with Donald's guidance: oxygen pressure, plume shape, target/substrate geometry, and temperature stability.
- Connecting growth choices to AFM, XRD, PFM, and electrical characterization results.

CONNECTED PROJECTS

- Nanosecond Pulse Generator: hardware for fast electrical characterization of oxide devices after fabrication.
- PLDTracker: data tool linking deposition conditions to characterization data and lab documentation.
- The larger loop is simple: grow films, track how they were grown, then measure how the devices switch.

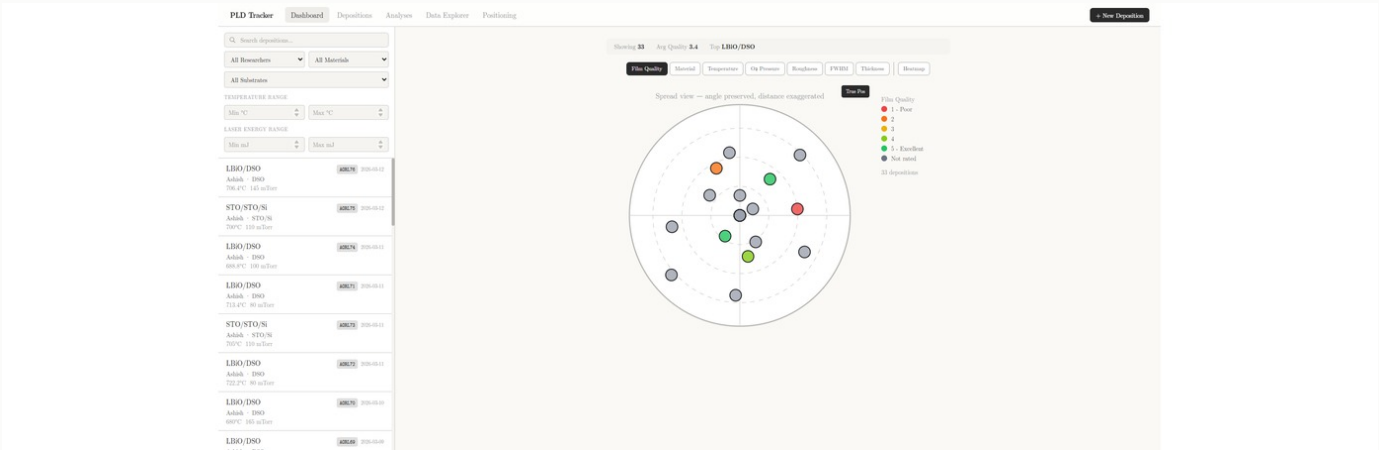
WHY IT MATTERS

Strain, interfaces, atomic stacking order, and oxygen vacancies can create switching behavior that does not exist in a single bulk material. I am mainly trying to understand how process choices show up later in device behavior.

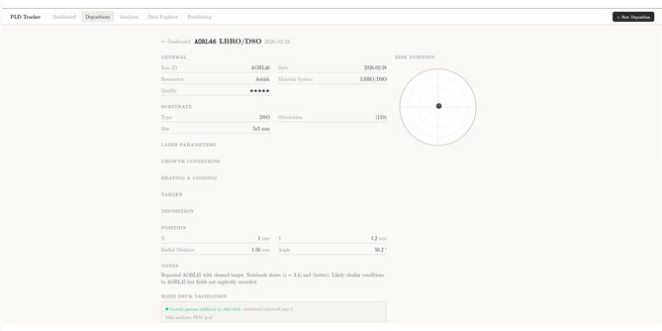
PLDTracker (Ramesh Lab)

Jan 2026 - Mar 2026 / Research Software

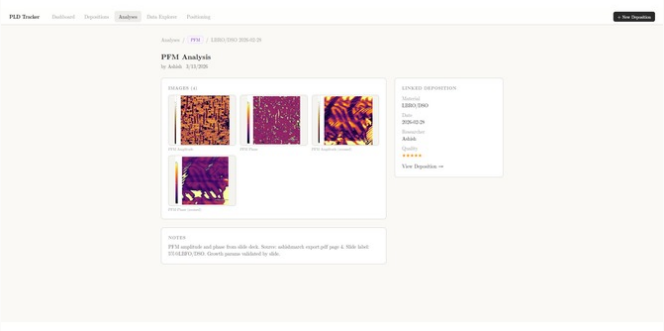
A lab tool for keeping PLD runs, wafer position, growth parameters, images, and slide-deck analysis from getting scattered across notes and folders.



Dashboard and deposition data explorer for tracking thin-film growth records.



Filtering and data exploration interface.



Analysis-image/document workflow tied back to deposition records.

IMPLEMENTED

- Logs substrate, target material, temperature, oxygen pressure, laser fluence, deposition time, and wafer disk position.
- Links XRD scans, AFM/PFM domain images, slide-deck figures, and camera captures to specific runs.
- Dashboard for filtering records and plotting parameter trends with connected datapoints and fit lines.

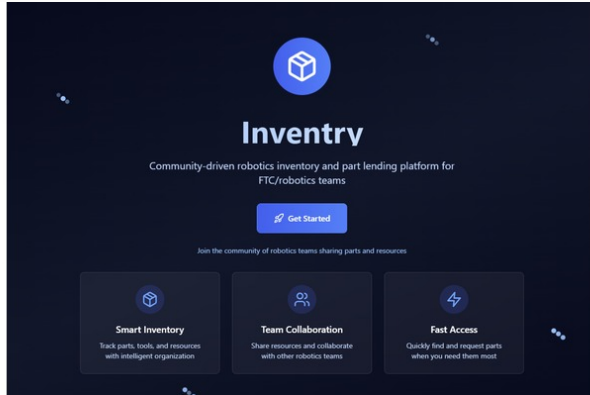
RESEARCH VALUE

- Replaces scattered notes, spreadsheets, screenshots, and slide decks with searchable experiment metadata.
- Makes it easier to ask whether a growth-condition change explains a film-structure or device-property change.
- Includes a SolidWorks camera mount path for substrate-position monitoring at the PLD chamber.

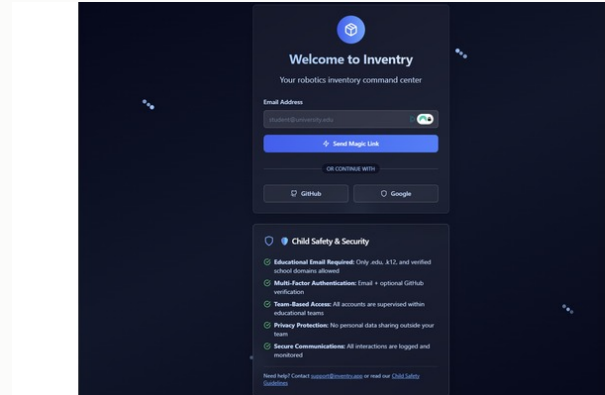
Inventory

Jun 2025 - Dec 2025 / Robotics Software

Robotics-parts inventory and lending system for FTC teams. It came from the very real problem of teams needing parts faster than shipping can deliver them.



Inventory landing/interface screenshot.



Inventory/product interface screenshot.

IMPLEMENTED

- Inventory management for parts, quantities, condition, team ownership, and search.
- Part lending network with distance matching, urgent requests, and lender reputation.
- Marketplace path for robotics components, with team verification through FIRST data.

TECHNICAL WORK

- React/TypeScript frontend, Node/Express backend, PostgreSQL data model, Vercel/Railway hosting.
- Invoice parsing with Tesseract OCR and GPT-4 structured extraction for part numbers, quantities, and prices.
- Weekly scrapers for GoBilda, REV, ServoCity, and AndyMark with 3,000+ SKU search data.

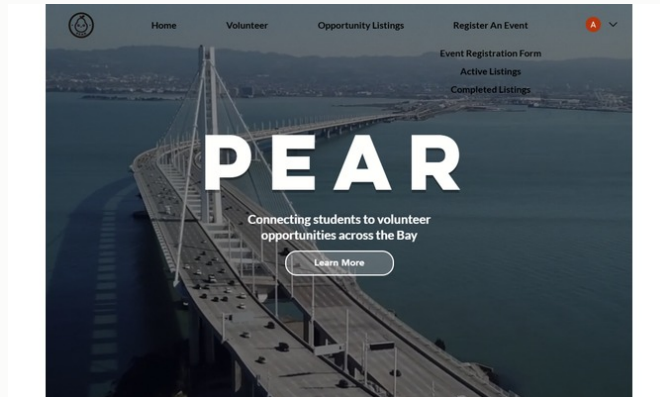
TOOLS

React / TypeScript / Node.js / Express / PostgreSQL / Tesseract OCR / Web Scraping / OAuth 2.0

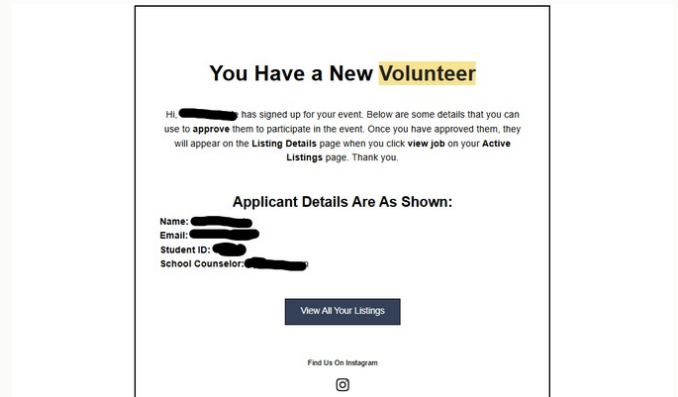
Pear Volunteering

Aug 2023 - May 2025 / Web Platform

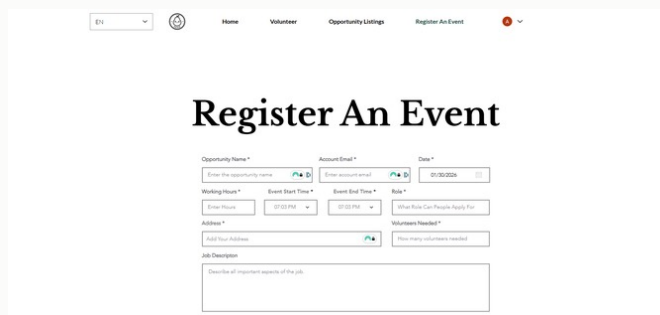
Volunteer platform for students, organizers, and administrators who needed event signup and verified hour logs to stop living in spreadsheets.



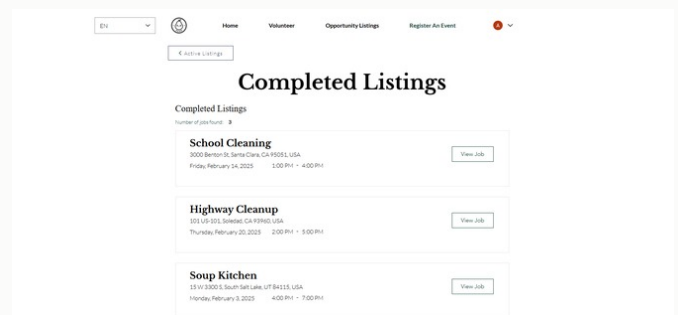
Pear volunteer/event platform screenshot.



Volunteer interface or organizer screen.



Supporting platform screenshot.



Volunteer workflow screenshot.

IMPLEMENTED

- Student event browsing, organizer event creation, administrator verification, and hour-log workflows.
- Student ID validation, schedule-conflict checks, capacity updates, organizer notifications, and QR check-in.
- Auto-generated verification letters as PDFs for documented service hours.

TECHNICAL WORK

- Wix Velo backend logic, custom JavaScript, webhook integrations, and role-based access.
- Async webhook queue for high-volume signup events that could fill 50+ spots quickly.
- Privacy constraint: student contact data stayed protected and communication flowed through the platform.

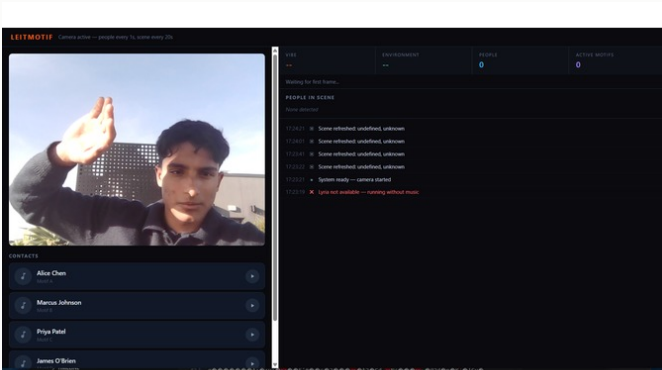
TOOLS

Wix Velo / JavaScript / Webhook Queues / QR Check-In / PDF Generation / Role-Based Access / Data Privacy

Leitmotif

Mar 2026 / Hackathon / Accessibility Software

YC x DeepMind hackathon winner. Real-time generative music interface that assigns musical motifs to people in a scene.



Leitmotif web/app interface screenshot.



Hackathon or demo context.

IMPLEMENTED

- Camera loop sends frames to Gemini vision, extracts scene/person state, then updates the music engine.
- Each person gets a persistent musical motif with entry and exit cues.
- Crossfades emotional-state changes over 4-6 seconds so scene updates do not sound abrupt.

ARCHITECTURE

- Two-server split: camera analysis and real-time music orchestration decoupled for different latency profiles.
- Gemini 2.5 Flash for vision/orchestration, Lyria RealTime for PCM audio generation, Supabase for persistence.
- React Native companion app with monitor, contacts, visualizer, and settings views.

TOOLS

TypeScript / Node.js / React Native / Gemini API / Lyria Realtime / Supabase / Computer Vision

Tools + Methods

Current / Engineering Toolkit

A compact map of the tools and methods that show up across the project pages.

CAD / MECHANICAL

- SolidWorks
- Onshape
- Fusion 360
- GD&T
- DFM/DFA
- FEA
- mechanism packaging
- tolerance stackups

MANUFACTURING

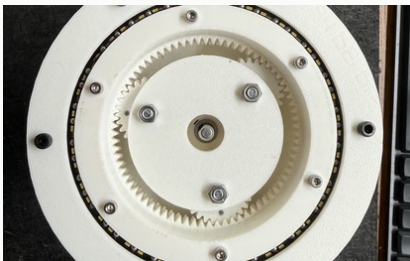
- mill
- lathe
- waterjet
- laser cutter
- TIG
- FDM/SLA
- Markforged
- fastener fit-up

ELECTRONICS / TEST

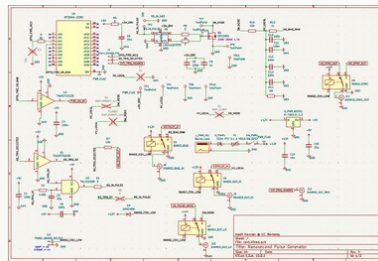
- KiCad
- LTspice
- oscilloscope
- source-measure
- RP2040/Python
- coax/BNC interfaces
- microcontrollers

SOFTWARE / DATA

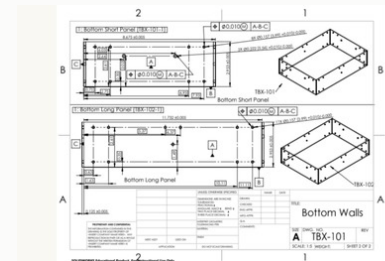
- Python
- MATLAB/Simulink
- Java
- React/TypeScript
- Node.js
- PostgreSQL
- JupyterLab
- LabVIEW



Actuator prototype



PCB schematic



GD&T drawing